

MSBTE Class Test-I Question Paper Solution

Program Name:- Artificial Intelligence and Machine Learning

Program Code:- AN

Course Title:- AI & ML Algorithm

Total Marks:- 30

Semester:- Fifth

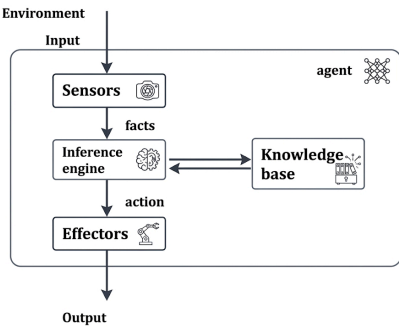
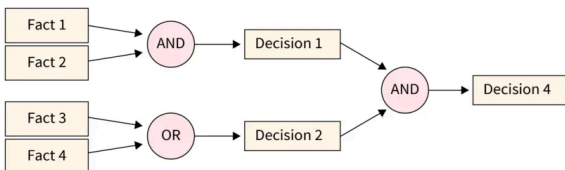
Course Code:- 315330

Time period:- 1.30 Hour

Instructions:-

- 1) Square bracket indicates course outcomes.
- 2) All Questions are Compulsory.
- 3) Assume suitable data, if necessary.

Que No.	Sub. Que. No.	Questions	Course Outcome	Blooms Level
1	Attempt any FIVE. (5 * 2 = 10 Marks)			
	a	Define Artificial Intelligence. Ans: Artificial Intelligence (AI) is the branch of computer science that enables machines to perform tasks requiring human intelligence. These tasks include learning, reasoning, problem-solving, and decision-making.	[315330.1]	Level 1
	b	Explain any two applications of AI. Ans: Healthcare: AI helps doctors diagnose diseases, analyze medical images, and recommend treatments. Virtual Assistants: AI-powered assistants like Siri and Google Assistant understand voice commands and answer user queries.	[315330.1]	Level 2
	c	State different types of AI agents. Ans: Simple Reflex Agent, Model-Based Reflex Agent, Goal-Based Agent, Utility-Based Agent, Learning Agent	[315330.1]	Level 1
	d	Explain the need of the Turing Test in AI. Ans: The Turing Test is used to determine whether a machine can exhibit intelligent behavior similar to a human. If a human cannot distinguish the machine from another human during conversation, the machine is considered intelligent.	[315330.1]	Level 2
	e	Identify the Causes of Uncertainty. Ans: Uncertainty in AI is caused by incomplete information, noisy or inaccurate data, unpredictable environments, and limitations in knowledge or observations.	[315330.2]	Level 3
	f	Define reasoning in AI. Ans: Reasoning in AI is the process of using available knowledge and logical rules to draw conclusions, make decisions, or solve problems.	[315330.2]	Level 1
	g	State the importance of Bayes' Theorem. Ans: Bayes' Theorem helps AI update the probability of an event when new evidence becomes available. It is widely used in prediction, decision-making, and machine learning.	[315330.2]	Level 3
2	Attempt any Five. (5 * 4 = 20 Marks)			
	a	Describe the scope and components of Artificial Intelligence. Ans: Scope of AI: i. AI is used to develop systems that can think, learn, and make decisions like humans. ii. It is applied in healthcare, robotics, education, finance, gaming, transportation, and cybersecurity. Components of AI: i. Learning – Acquiring knowledge from data.	[315330.1]	Level 1

		ii. Reasoning – Making logical decisions. iii. Problem Solving – Finding solutions to complex problems. iv. Knowledge Representation – Storing facts and information. v. Perception – Understanding images, speech, and surroundings. vi. Natural Language Processing (NLP) – Understanding and generating human language.		
	b	Explain types of intelligent agents with suitable examples. Ans: i. Simple Reflex Agent – Acts based on the current situation only. Example: Automatic door opens when someone approaches. ii. Model-Based Reflex Agent – Uses memory of previous states. Example: Robot vacuum remembers cleaned areas. iii. Goal-Based Agent – Chooses actions to achieve a specific goal. Example: GPS navigation finds the shortest route. iv. Utility-Based Agent – Selects the best action based on usefulness. Example: Self-driving car choosing the safest and fastest path. v. Learning Agent – Improves performance through experience. Example: Netflix recommending movies based on viewing history.	[315330.1]	Level 2
	c	State search algorithm and list its properties. Ans: A search algorithm is a method used by AI to find a path or solution from an initial state to a goal state by exploring possible alternatives. Properties of Search Algorithms: i. Completeness: Finds a solution if one exists. ii. Optimality: Finds the best solution. iii. Time Complexity: Time taken to find the solution. iv. Space Complexity: Memory required during the search.	[315330.1]	Level 1
	d	Apply Generate-and-Test technique to solve a problem. Ans: It generates possible solutions one by one and tests each until the correct solution is found. Example: Find x such that $x^2 = 16$. Generate $x = 1 \rightarrow 1^2 = 1$ Generate $x = 2 \rightarrow 2^2 = 4$ Generate $x = 3 \rightarrow 3^2 = 9$ Generate $x = 4 \rightarrow 4^2 = 16$ Result: The solution is $x = 4$.	[315330.1]	Level 3
	e	Explain architecture of Knowledge-Based Agent with diagram. Ans:  Sensors: Collect information. Knowledge Base: Stores facts and rules. Inference Engine: Applies logical reasoning. Actuators: Perform actions.	[315330.2]	Level 2
	f	Explain Forward Chaining & Backward Chaining with diagrams. Ans: i. Forward Chaining  Starts with known facts & applies rules until a conclusion is reached. It is data-driven. ii. Backward Chaining a	[315330.2]	Level 2

		Starts with the goal and works backward to verify whether the facts support it. It is goal-driven.		
	g	Apply Bayes' Theorem to find probability of simple condition. Ans: $P(A B) = \frac{P(B A) P(A)}{P(B)}$ i. A disease affects 2% of people. A test correctly detects the disease 95% of the time, and gives a false positive 5% of the time. ii. Using Bayes' Theorem: P(Disease) = 0.02 P(Positive Disease) = 0.95 P(Positive) = (0.95 × 0.02) + (0.05 × 0.98) = 0.068 iii. Therefore, P(Disease Positive) = (0.95 × 0.02) / 0.068 ≈ 0.279 (27.9%) iv. Even after testing positive, the probability of actually having the disease is about 28% because the disease is rare and false positives occur.	[315330.2]	Level 3